

VINAY JOSHI Ph.D. | AI Research Scientist

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BIO

AI Research Scientist (Ph.D.) with deep expertise in applications of **large language models**, their architecture, training, inference, and evaluations. Strong record of **publications, patents**, and leading collaborative AI projects with global teams. **8+ years** experience in **Python, PyTorch** including **3+ years** of leadership experience on generative AI projects.

WORK EXPERIENCE

Microsoft Bangalore, IN
Principal Applied Scientist Oct 2025 - Present

- **Leadership:** Currently leading high-impact initiatives for **Microsoft 365 Copilot**, specifically driving AI-enabled search relevance for **Teams** by **managing cross-functional** technical workflows and improving LLM trustworthiness and hallucination detection.
- **Talent Orchestration & Growth:** Actively scaling the Applied Sciences organization in Bangalore by mentoring a diverse group of FTEs to accelerate the delivery of M365 Copilot features, while steering the regional hiring strategy to secure and elevate top-tier AI talent.

AMD Bangalore, IN
Member of technical staff/Tech lead Dec 2024 - Oct 2025

- **AI for productivity:** Led a team in developing GenAI solution for developer productivity enhancement for GPU kernel generation using advanced agentic approaches, fine-tuning and alignment of LLMs.
- **Leadership:** Mentored FTEs and managing interns across GenAI projects on code generation.
- **GenAI optimization:** Contributed to generative pretraining of efficient and lean foundational large language models for broader use cases.

Intel Labs Bangalore, IN
AI Research Scientist Dec 2021 - Dec 2024

- **Lead AI Optimization Research:** Led a team to develop novel methods for optimizing large language models (LLMs), including the QCQA framework, which significantly improves KV-cache efficiency without compromising accuracy. This innovation is under review for technology transfer within Intel.
- **DLRM optimization research:** Led a team of 2 Engineers to develop a technique for optimizing the allocation of large embedding data in DLRM to enhance performance on memory/bandwidth-constrained systems.
- **Mentorship and Collaboration:** Regularly mentor junior researchers and collaborate with teams globally on high-impact AI projects.
- **LLM and AI Workload Profiling:** Conducted in-depth profiling of SOTA AI models on Intel's CPU and GPU to address bottlenecks and propose improvements.

IBM Research Labs Zurich Zurich, Switzerland
Visiting Research Scientist (enrolled in NJIT, USA for PhD) Jun 2018 - Dec 2019

- **AI for Hardware Acceleration:** Played a key role in the IBM AI challenge, developing simulators in TensorFlow to evaluate novel hardware solutions for large-scale machine learning training.

Mahindra Susten PVT. LTD. Mumbai Mumbai, India
Engineer | Special Projects Division Jul 2016 - Jul 2017

EDUCATION

King's College London (enrolled in NJIT, USA for 2017-20 then transferred to KCL) London, UK
Ph.D. Department of Natural & Mathematical Sciences Apr 2020 - Jun 2022

Indian Institute of Technology Bombay Mumbai, India
M.Tech. Electrical Engineering *GPA: 9.2/10* Aug 2013 - Jul 2016

University of Pune Pune, India
B.E. in E&TC Engineering *GPA: First class with distinction* Aug 2009 - Jun 2013

TECHNICAL SKILLS

Frameworks:	Numpy, PyTorch, JAX, TorchTune, Huggingface, TensorFlow, PyMoo
Programming languages:	Python, C, C++
Container tools:	Docker, Singularity
Development tools:	Git, Bash, Python debugger, Amazon web services, Slurm, Firebase
Low level programming:	Triton, torch.compile
AI model Serving tools:	VLLM, SGLang, Torch serve

NOTABLE GENAI AND RESEARCH PROJECTS

Total citations (as of June 1, 2026): **1,396** | h-index: **9** | i10-index: **9**

Jianghui Wang, **Vinay Joshi**, et al.,

GEAK: Introducing Triton Kernel AI Agent & Evaluation Benchmarks,

ArXiv article, **ROCm technical blog**

Vinay Joshi, et al.,

TaDA: Training-free recipe for Decoding with Adaptive KV Cache Compression and Mean-centering,

ACL Industry track 2025

Vinay Joshi, et al.,

QCQA: Quality and Capacity-aware grouped Query Attention,

Preprint - under review at NeurIPS 2024, doi - <https://arxiv.org/abs/2406.10247>

Vinay Joshi, Wangxin He, Jae-sun Seo, Bipin Rajendran

Hybrid In-memory Computing Architecture for the Training of Deep Neural Networks, **International Symposium on Circuits and Systems 2021**.

Vinay Joshi, et al.,

Accurate deep neural network inference using computational phase-change memory,

Nature Communications 2020, doi - <https://doi.org/10.1038/s41467-020-16108-9>

AWARDS

Exceptional Research Lead Award | May 2024 | Intel Labs

This award is in recognition of producing novel ideas, leading research teams, and self-motivated successful patent filing, publication, and technology transfer initiation.

IBM Research Leadership Award | Apr 2019 | IBM, USA.

This award is in recognition of the successful filing of the **patents** and **accepted publication** by **leading and spearheading** ideas at IBM.

Provost Doctoral Assistantship Award | Sep 2017 | NJIT, NJ, USA.

Awarded to the **top 10 incoming graduate students** at NJIT.

Winner of the Ideathon contest | Jan 2015 | VLSID & ES conference, Bangalore, India.

Developed a prototype in **60 hours** to sense mosquito density in a locality with **IoT infrastructure**.

Best Speaker Award | Jul 2016 | Reading group, IIT Bombay

I delivered an oral presentation on **solving real-world problems with data and deep learning**.

INVENTIONS

— **Granted**

Electronic system for computing items of an outer product matrix | US Patent Application no.:16435248, Date:2020/12/10. Co-inventors: Abu Sebastian, Manuel Le Gallo, Christophe Piveatue, Irem Boybat-kara.

Training of artificial neural networks | US Patent Application no.:16380312, Date:2020/10/15. Co-inventors: Abu Sebastian, Manuel Le Gallo, Christophe Piveatue.

Conductance drift corrections in neuromorphic systems based on crossbar array structures | US Patent Application no.:16842874, Date:2021/10/14. Co-inventors: Abu Sebastian, Manuel Le Gallo, Simon Haefeli.

Gated unit for a gated recurrent neural network | US Patent Application no.:17074797, Date:2022/04/21. Co-inventors: Abu Sebastian, Manuel Le Gallo, Milos Stanisavljevic.

— **Recently filed**

QCQA: Quality and Capacity-aware grouped query attention | Date:2024/06/06. Co-inventors: Prashant Laddha, Om Ji Omer.

Streaming N-dimensional Matrix Transpose (Tensor Reshape) Hardware for TMUL like GEMM engines with an ISA | Date:2023/08/28. Co-inventors: Kamlesh Pillai, Om Ji Omer.

LANGUAGES

I have professional proficiency in the English language and native proficiency in Hindi and Marathi.

HOBBIES

During my spare time, I like to read, go ice skating, play volleyball, go swimming, or play chess.